In [29]:
```
# step1 : we will import the libraries

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style="whitegrid")
plt.rcParams["figure.figsize"] = (8,5)
```

In [30]:
```
# step2: load the datasets

df = pd.read_csv("C:/Users/sarvesh/Downloads/student_data.csv")
df.head()
```

Out[30]:

	school	sex	age	address	famsize	Pstatus	Medu	Fedu	Mjob	Fjob	...	famrel	freetime
0	GP	F	18	U	GT3	A	4	4	at_home	teacher	...	4	3
1	GP	F	17	U	GT3	T	1	1	at_home	other	...	5	3
2	GP	F	15	U	LE3	T	1	1	at_home	other	...	4	3
3	GP	F	15	U	GT3	T	4	2	health	services	...	3	2
4	GP	F	16	U	GT3	T	3	3	other	other	...	4	3

5 rows × 33 columns

In [31]:
```
# step3 : we will understand the basics knowledge
```

In [32]:
```
df.shape
```

Out[32]:
```
(395, 33)
```

In [33]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 395 entries, 0 to 394
Data columns (total 33 columns):
 #   Column      Non-Null Count  Dtype 
---  ------      --------------  ----- 
 0   school      395 non-null    object
 1   sex         395 non-null    object
 2   age         395 non-null    int64 
 3   address     395 non-null    object
 4   famsize     395 non-null    object
 5   Pstatus     395 non-null    object
 6   Medu        395 non-null    int64 
 7   Fedu        395 non-null    int64 
 8   Mjob        395 non-null    object
 9   Fjob        395 non-null    object
 10  reason      395 non-null    object
 11  guardian    395 non-null    object
 12  traveltime  395 non-null    int64 
 13  studytime   395 non-null    int64 
 14  failures    395 non-null    int64 
 15  schoolsup   395 non-null    object
 16  famsup      395 non-null    object
 17  paid        395 non-null    object
 18  activities  395 non-null    object
 19  nursery     395 non-null    object
 20  higher      395 non-null    object
 21  internet    395 non-null    object
 22  romantic    395 non-null    object
 23  famrel      395 non-null    int64 
 24  freetime    395 non-null    int64 
 25  goout       395 non-null    int64 
 26  Dalc        395 non-null    int64 
 27  Walc        395 non-null    int64 
 28  health      395 non-null    int64 
 29  absences    395 non-null    int64 
 30  G1          395 non-null    int64 
 31  G2          395 non-null    int64 
 32  G3          395 non-null    int64 
dtypes: int64(16), object(17)
memory usage: 102.0+ KB
```

In [34]:
```
df.describe()
```

Out[34]:

	age	Medu	Fedu	traveltime	studytime	failures	famrel	freetime
count	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000
mean	16.696203	2.749367	2.521519	1.448101	2.035443	0.334177	3.944304	3.235443
std	1.276043	1.094735	1.088201	0.697505	0.839240	0.743651	0.896659	0.998862
min	15.000000	0.000000	0.000000	1.000000	1.000000	0.000000	1.000000	1.000000
25%	16.000000	2.000000	2.000000	1.000000	1.000000	0.000000	4.000000	3.000000
50%	17.000000	3.000000	2.000000	1.000000	2.000000	0.000000	4.000000	3.000000
75%	18.000000	4.000000	3.000000	2.000000	2.000000	0.000000	5.000000	4.000000
max	22.000000	4.000000	4.000000	4.000000	4.000000	3.000000	5.000000	5.000000

In [35]:
```
# step4 : Data Cleaning
```

```
df.isnull().sum()
```

Out[35]:

```
school        0
sex           0
age           0
address       0
famsize       0
Pstatus       0
Medu          0
Fedu          0
Mjob          0
Fjob          0
reason        0
guardian      0
traveltime    0
studytime     0
failures      0
schoolsup     0
famsup        0
paid          0
activities    0
nursery       0
higher        0
internet      0
romantic      0
famrel        0
freetime      0
goout         0
Dalc          0
Walc          0
health        0
absences      0
G1            0
G2            0
G3            0
dtype: int64
```

In [36]:

```
df.columns = df.columns.str.replace(' ', '_')
df.head()
```

Out[36]:

	school	sex	age	address	famsize	Pstatus	Medu	Fedu	Mjob	Fjob	...	famrel	freetime
0	GP	F	18	U	GT3	A	4	4	at_home	teacher	...	4	3
1	GP	F	17	U	GT3	T	1	1	at_home	other	...	5	3
2	GP	F	15	U	LE3	T	1	1	at_home	other	...	4	3
3	GP	F	15	U	GT3	T	4	2	health	services	...	3	2
4	GP	F	16	U	GT3	T	3	3	other	other	...	4	3

5 rows × 33 columns

In [37]:

```
# step5 : here, we will understand the target variables
# G1 - First period grade
# G2 - Second period grade
# G3 - Final grade(Main Target)

df[['G1','G2','G3']].head()
```

Out[37]:

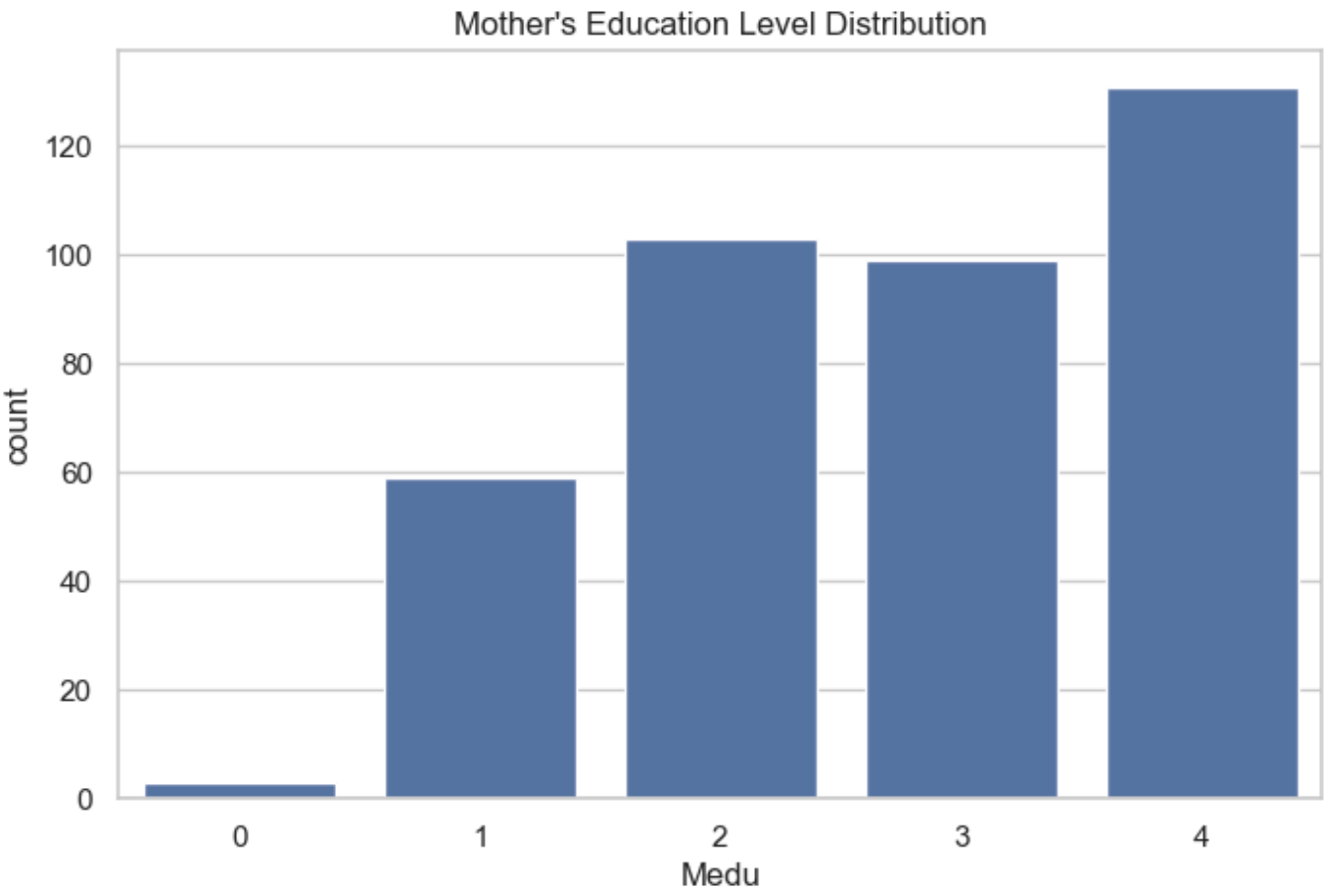
	G1	G2	G3
0	5	6	6
1	5	5	6
2	7	8	10
3	15	14	15
4	6	10	10

In [38]:
```
# step6 : create total_score and average_score

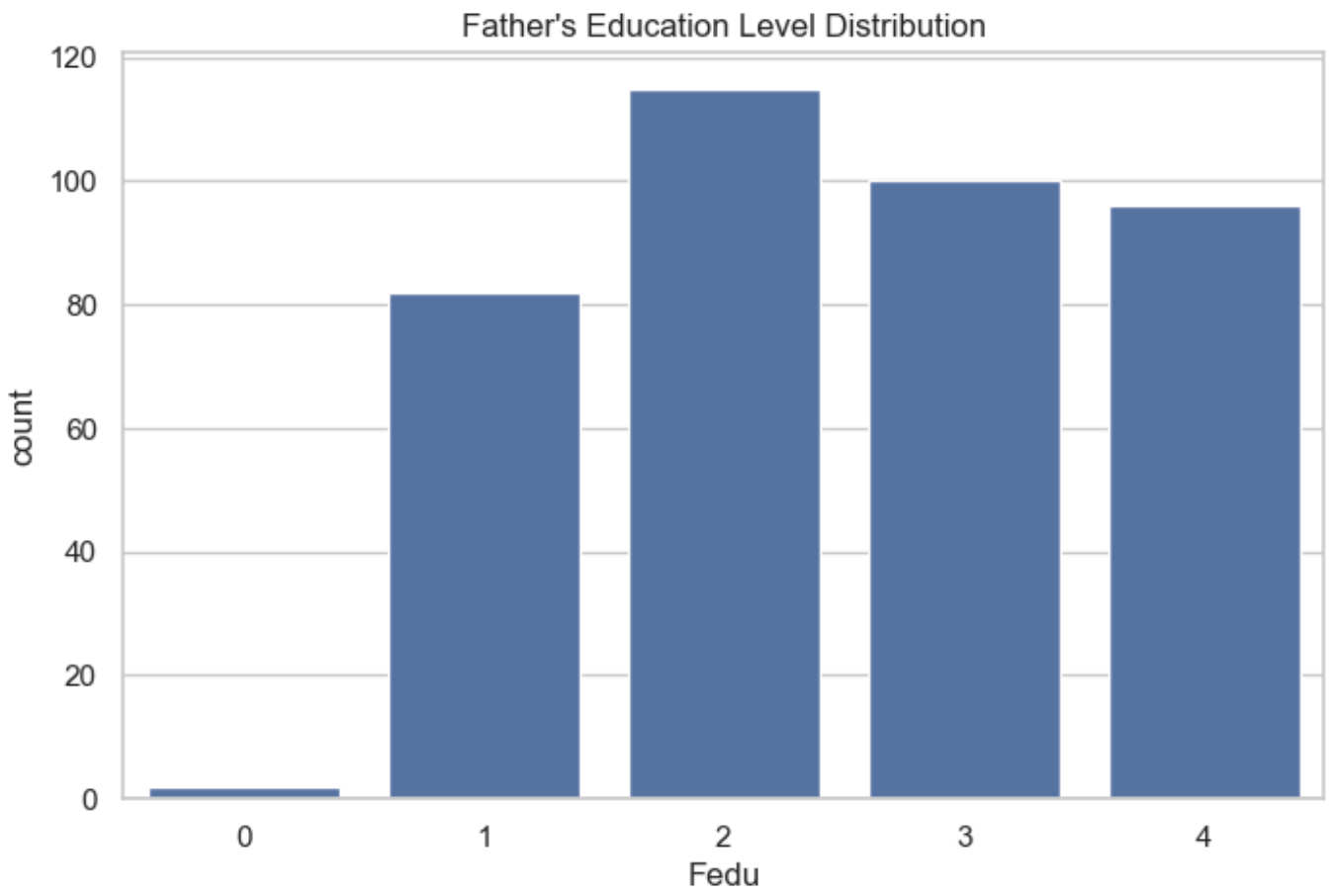
df['total_score'] = df['G1'] + df['G2'] + df['G3']
df['average_score'] = df['total_score'] / 3
```

In [39]:
```
# step7 : categorical feature distribution (we will check parent's education)
```

In [40]:
```
sns.countplot(x='Medu', data=df)
plt.title("Mother's Education Level Distribution")
plt.show()
```

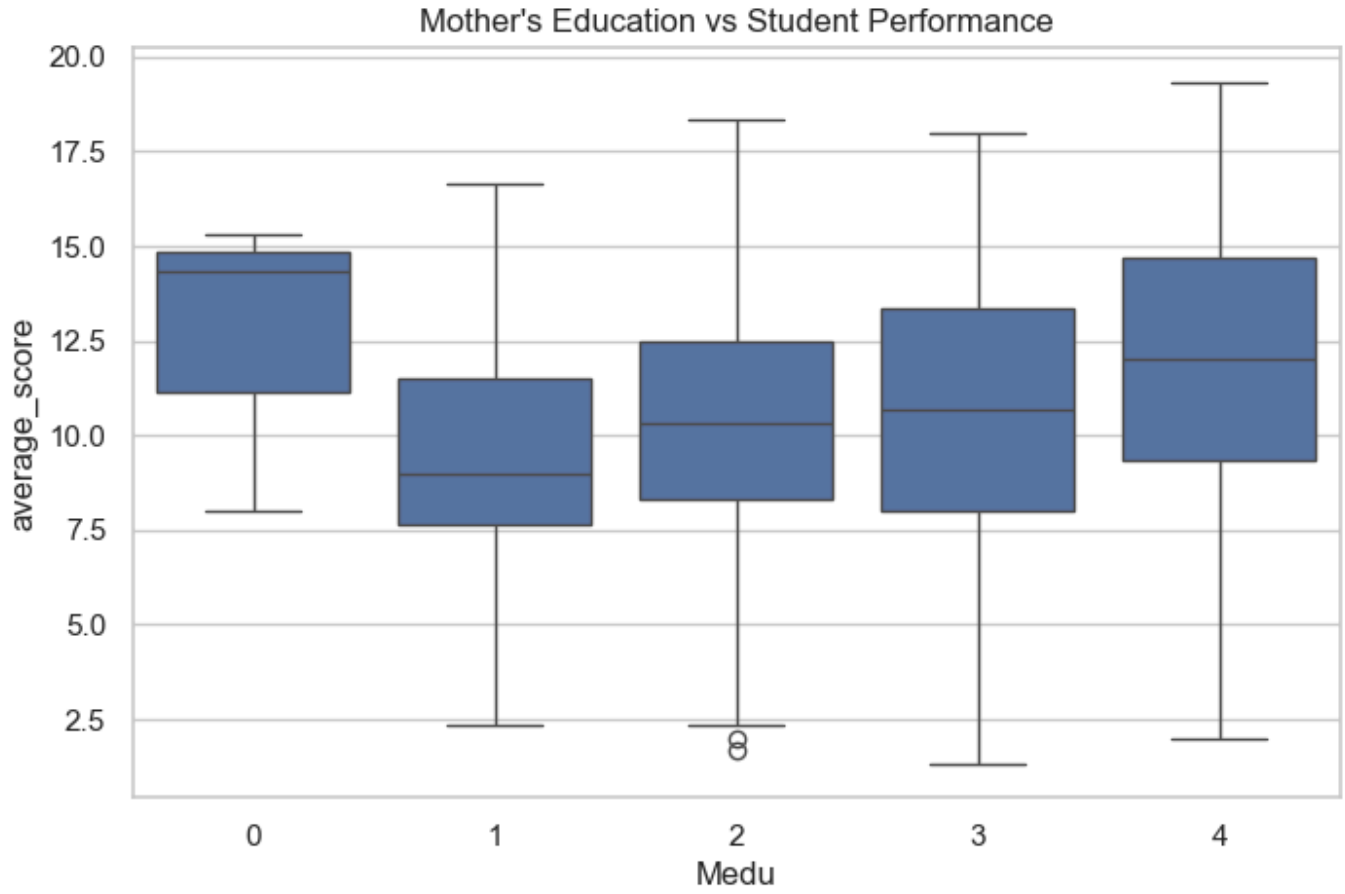


In [41]:
```
sns.countplot(x='Fedu', data=df)
plt.title("Father's Education Level Distribution")
plt.show()
```

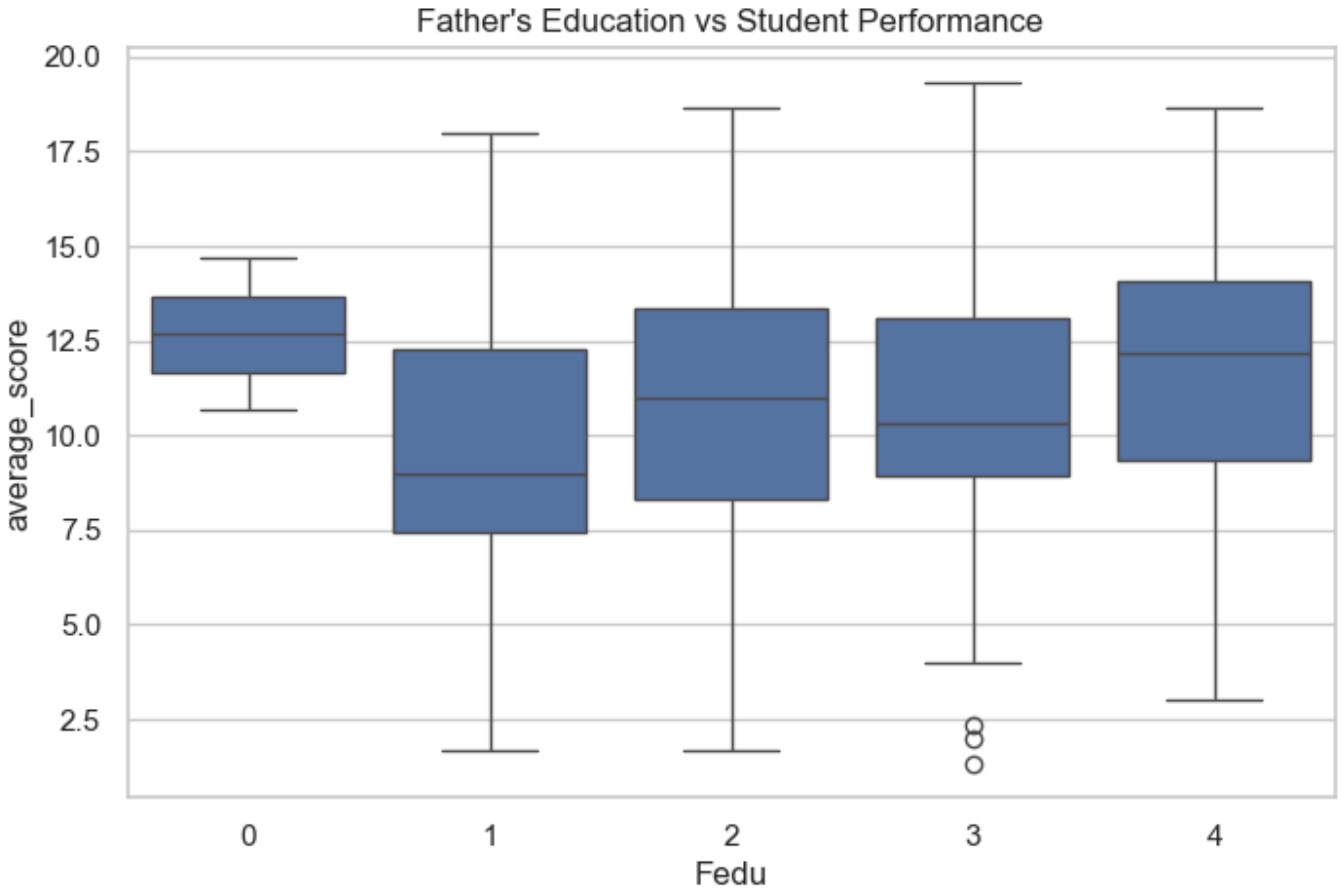


In [42]:
```
# step8 : Parent's Education vs Performance
```

In [43]:
```
sns.boxplot(x='Medu', y='average_score', data=df)
plt.title("Mother's Education vs Student Performance")
plt.show()
```

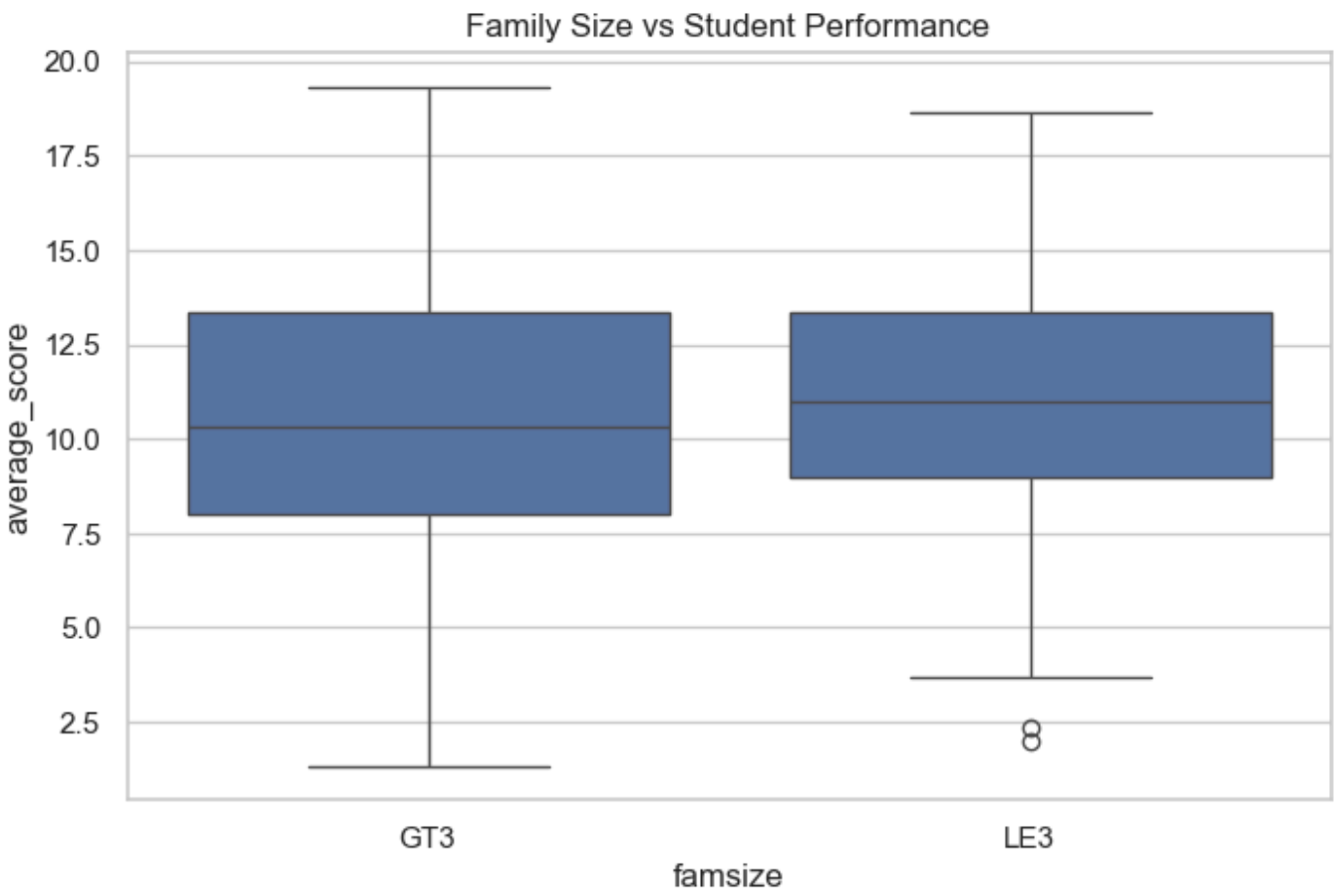


In [44]:
```
sns.boxplot(x='Fedu', y='average_score', data=df)
plt.title("Father's Education vs Student Performance")
plt.show()
```



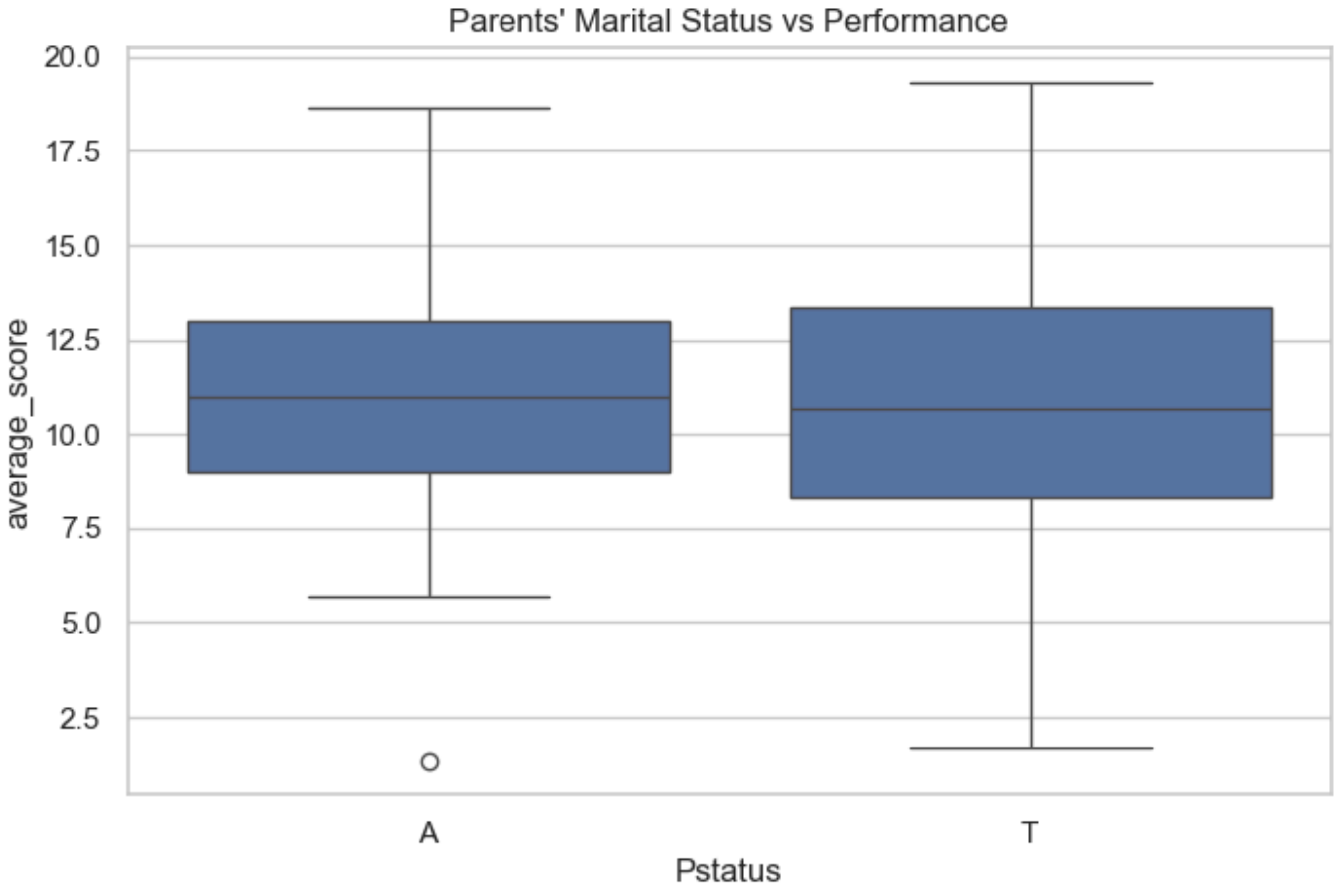
In [45]:
```
# step9 : we will check Family size impact

sns.boxplot(x='famsize', y='average_score', data=df)
plt.title("Family Size vs Student Performance")
plt.show()
```



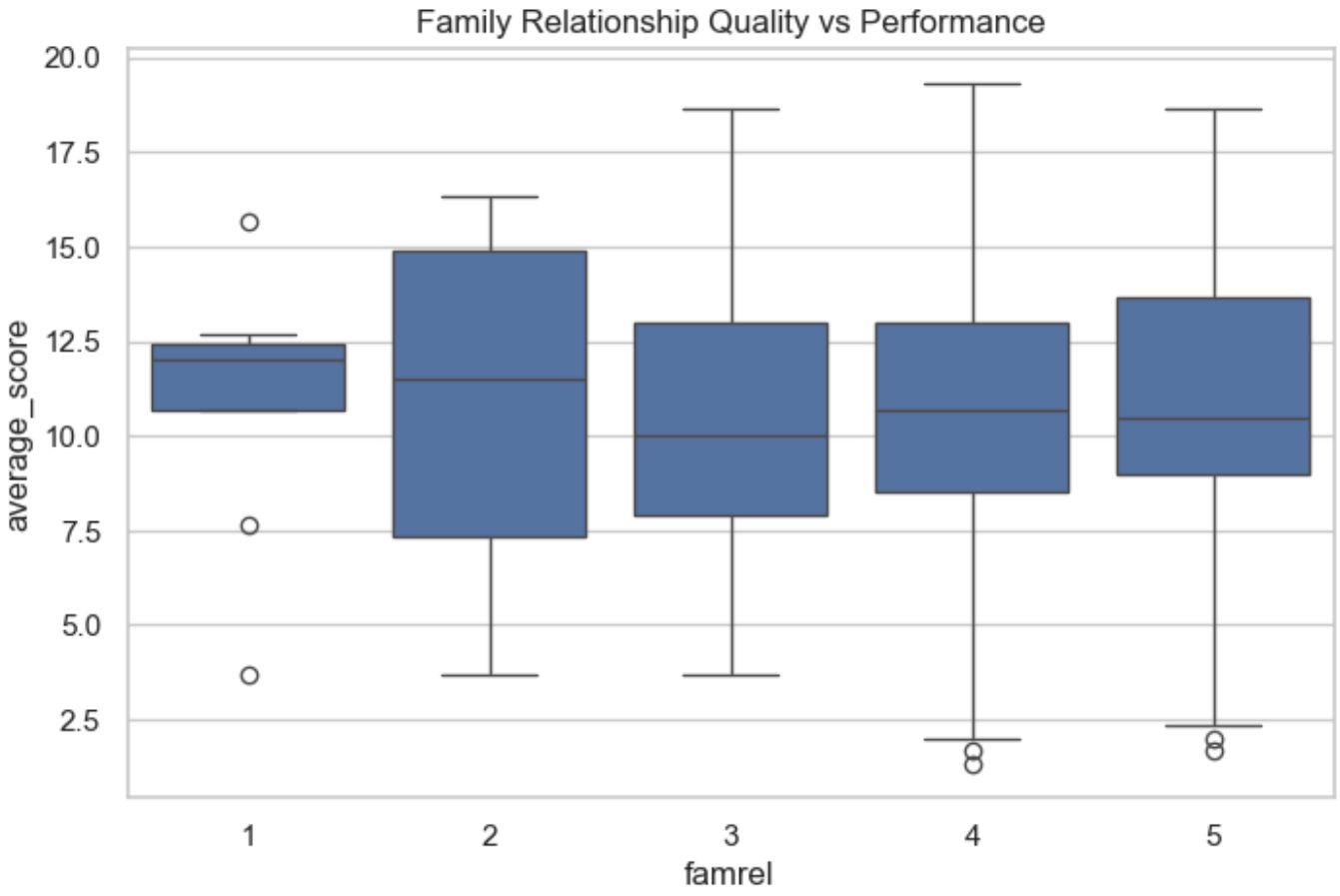
In [46]:
```
# step10 : Parent's Marital Status

sns.boxplot(x='Pstatus', y='average_score', data=df)
plt.title("Parents' Marital Status vs Performance")
plt.show()
```



In [47]:
```
# step11 : Family Relationship Quality

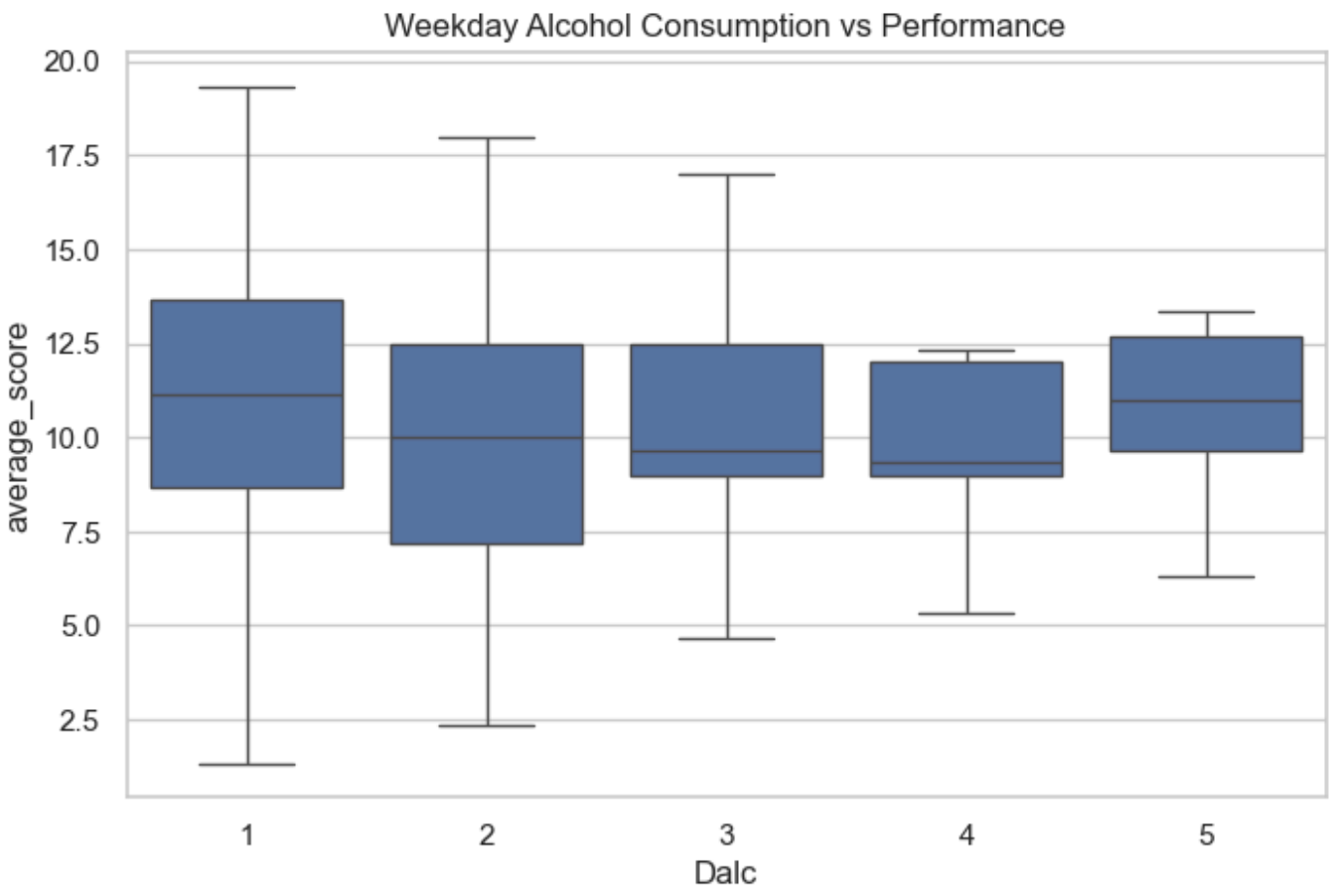
sns.boxplot(x='famrel', y='average_score', data=df)
plt.title("Family Relationship Quality vs Performance")
plt.show()
```



In [48]:
```
# step12: Alcohol Consumption Analysis
```

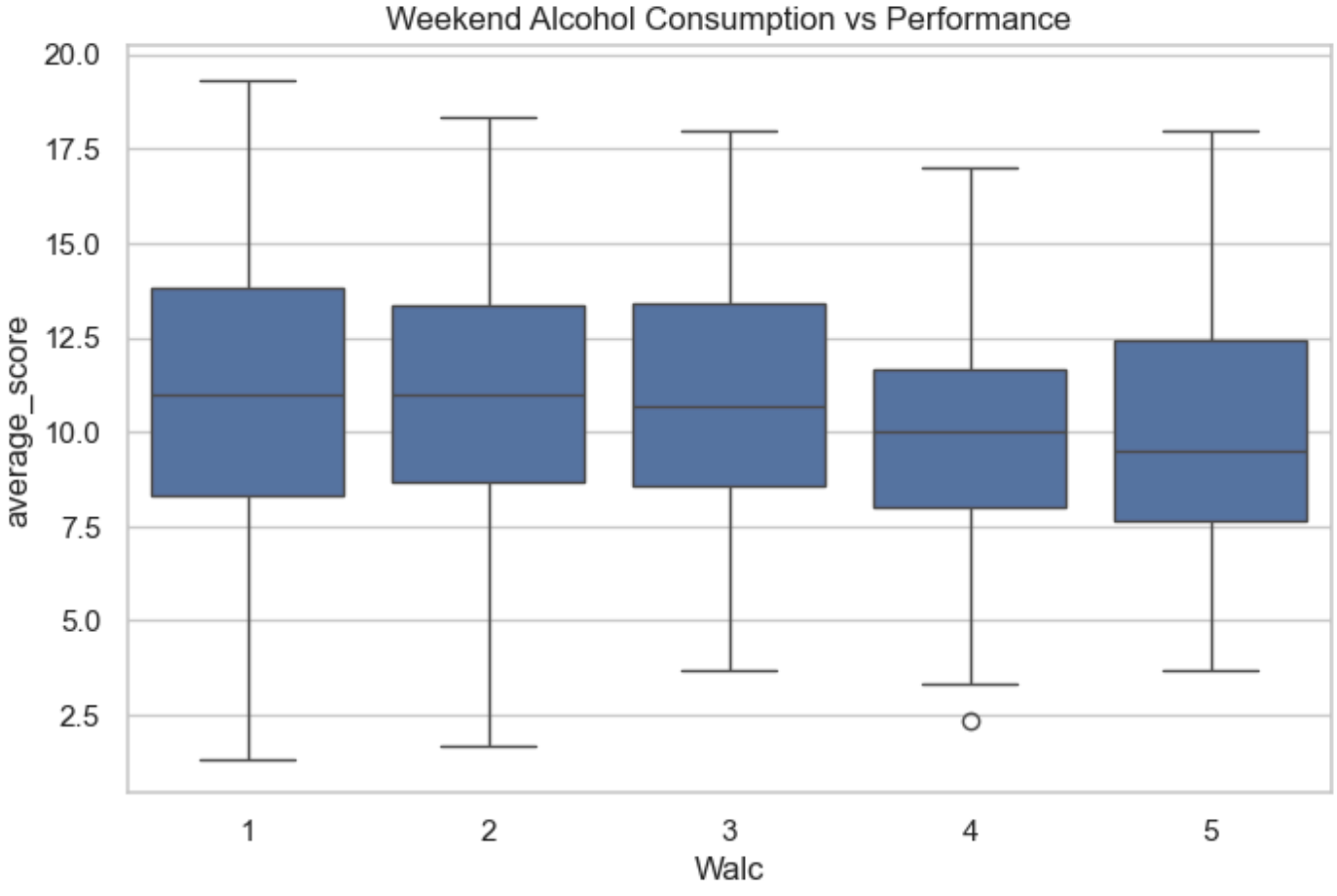
In [49]:
```
# here we will see Weekday Alcohol Consumption

sns.boxplot(x='Dalc', y='average_score', data=df)
plt.title("Weekday Alcohol Consumption vs Performance")
plt.show()
```



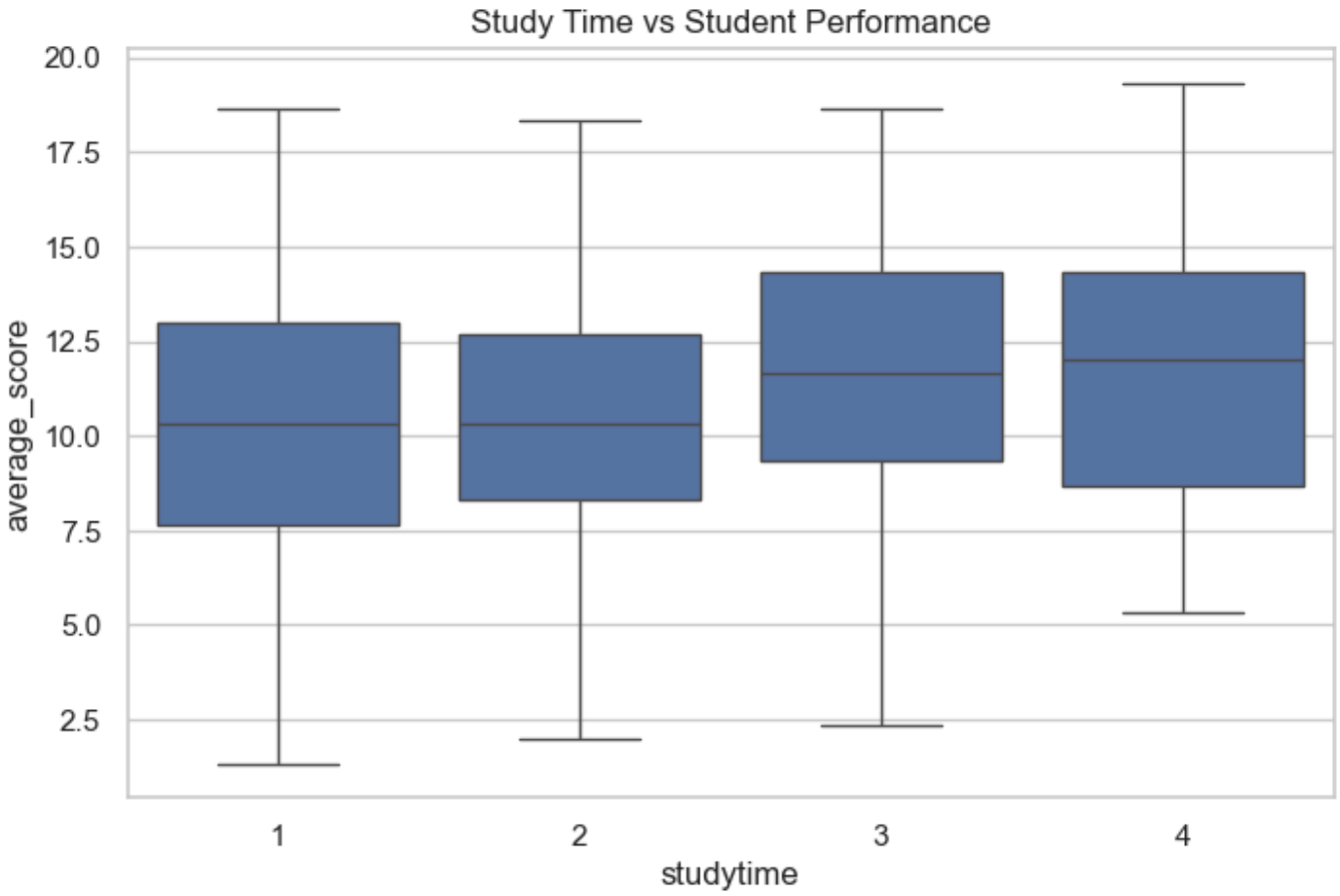
In [50]:
```
# here, Weekend Alcohol Consumption

sns.boxplot(x='Walc', y='average_score', data=df)
plt.title("Weekend Alcohol Consumption vs Performance")
plt.show()
```



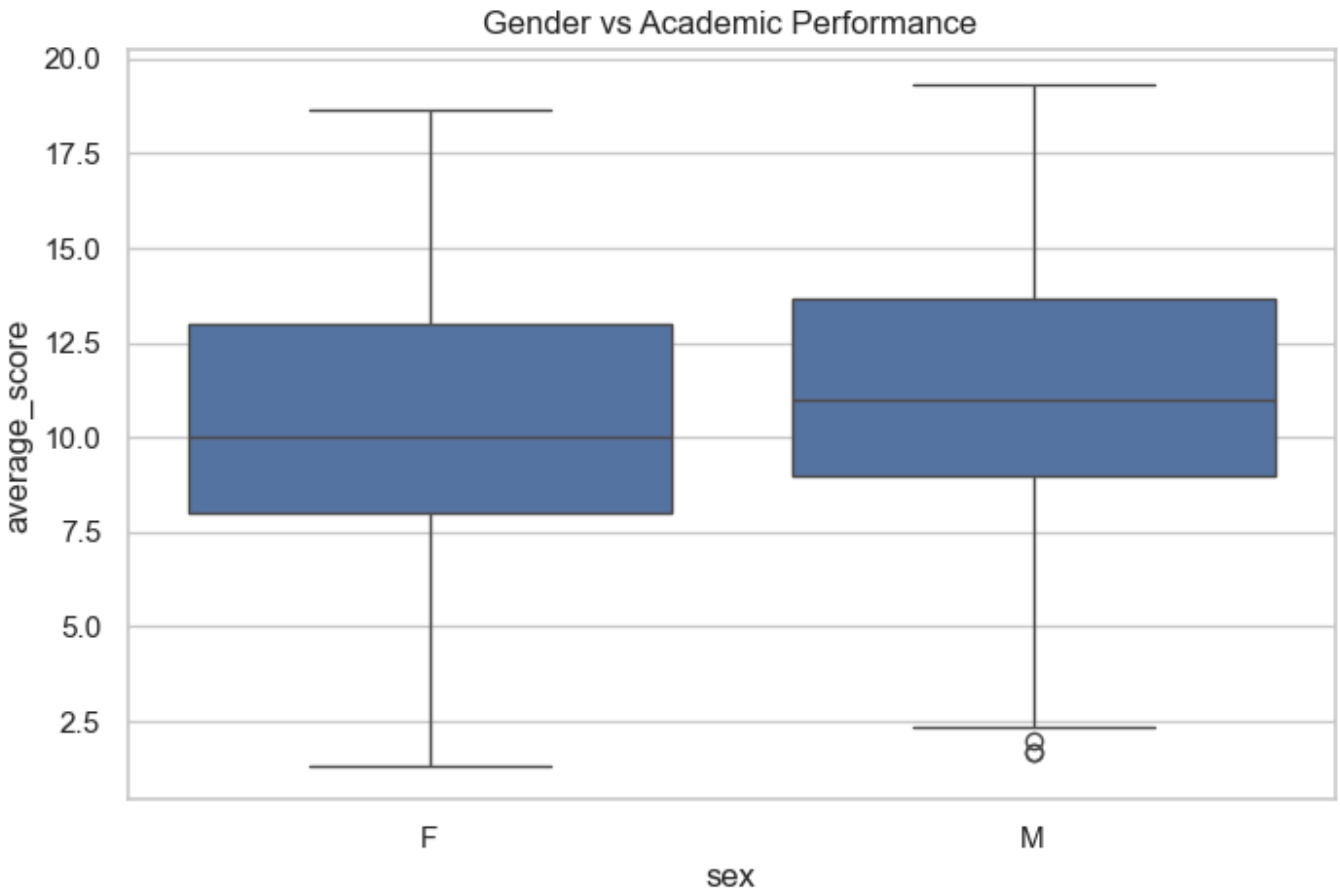
In [51]:
```
# step13 : Study Time vs Performance

sns.boxplot(x='studytime', y='average_score', data=df)
plt.title("Study Time vs Student Performance")
plt.show()
```



In [52]:
```
# step14 : Gender vs Performance

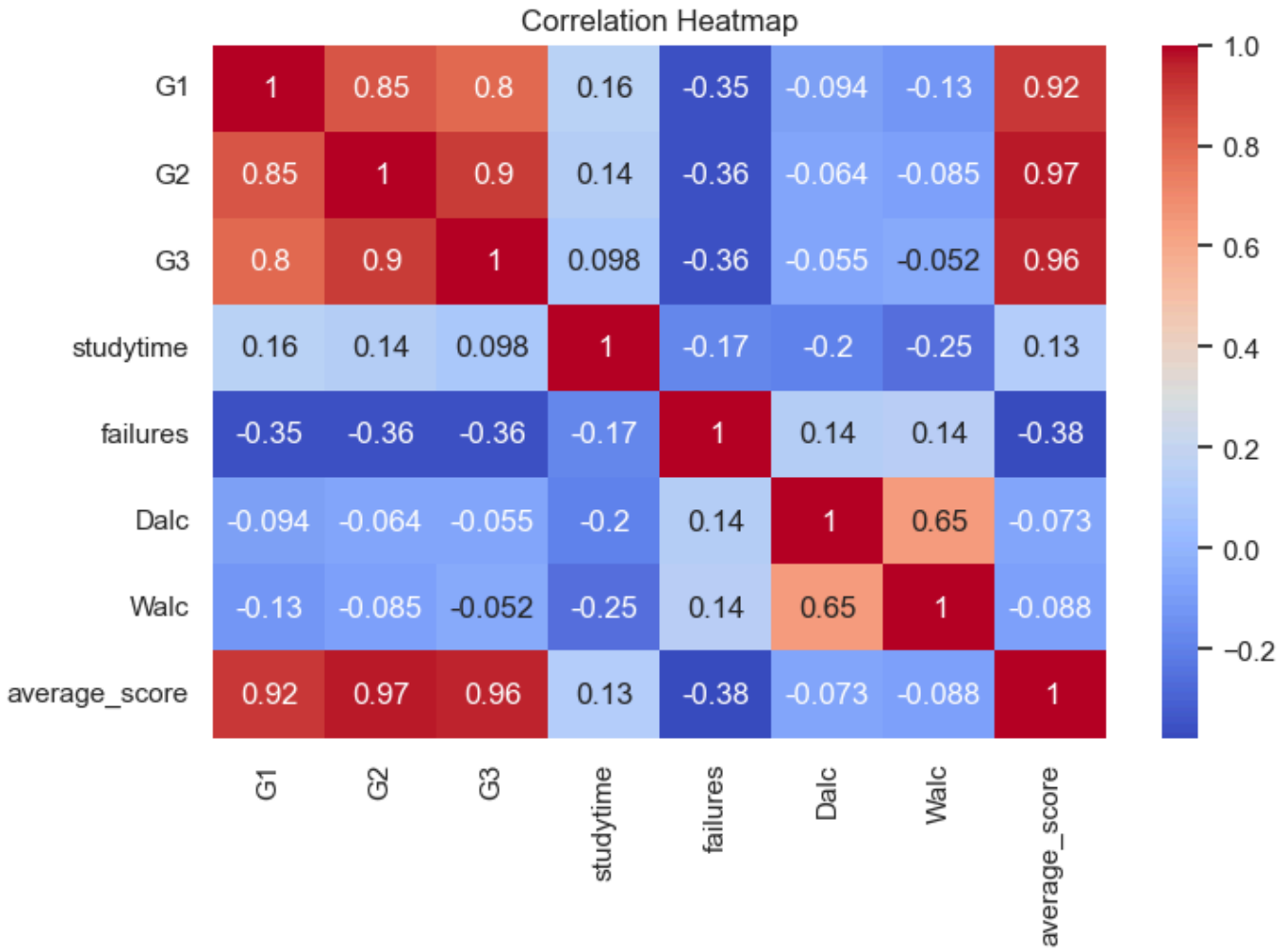
sns.boxplot(x='sex', y='average_score', data=df)
plt.title("Gender vs Academic Performance")
plt.show()
```



In [53]:
```
# step15 : here we'll do the Correlation Analysis

corr = df[['G1','G2','G3','studytime','failures','Dalc','Walc','average_score']].corr()

sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```



In [54]:
```
# step16 : Which Grade Impacts Final Score Most? (we can judge from the output)

df[['G1','G2','G3']].corr()
```

Out[54]:

	G1	G2	G3
G1	1.000000	0.852118	0.801468
G2	0.852118	1.000000	0.904868
G3	0.801468	0.904868	1.000000

In [ ]: